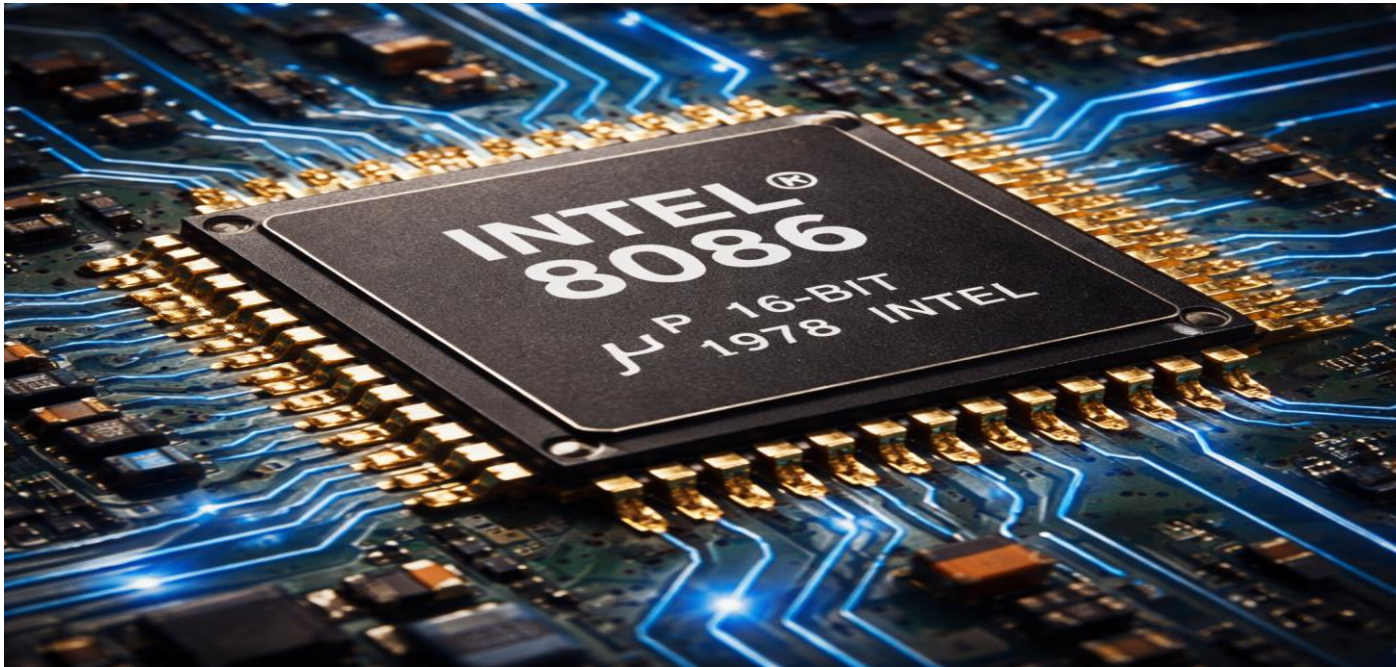


8086 Microprocessor Internal Registers



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8086 Microprocessor Internal Registers

8086 consists Of 13 registers, all of 16 bits size.

- Data Registers (AX, BX, CX, and DX).
- Segment Registers (CS, DS, SS, and ES).
- Instruction Pointer (IP).
- Index Registers (SI and DI).
- Pointer Registers (BP and SP).
- Status Register (SR) or (Flag Register).

General Purpose Registers (Data Registers) (AX, BX, CX, and DX)

AX Register (Accumulator)

Used for all Input/ Output operations, some string operations and Arithmetic operations.

BX Register (Base register)

Used as an index to extend addressing, it's also used for Computation.

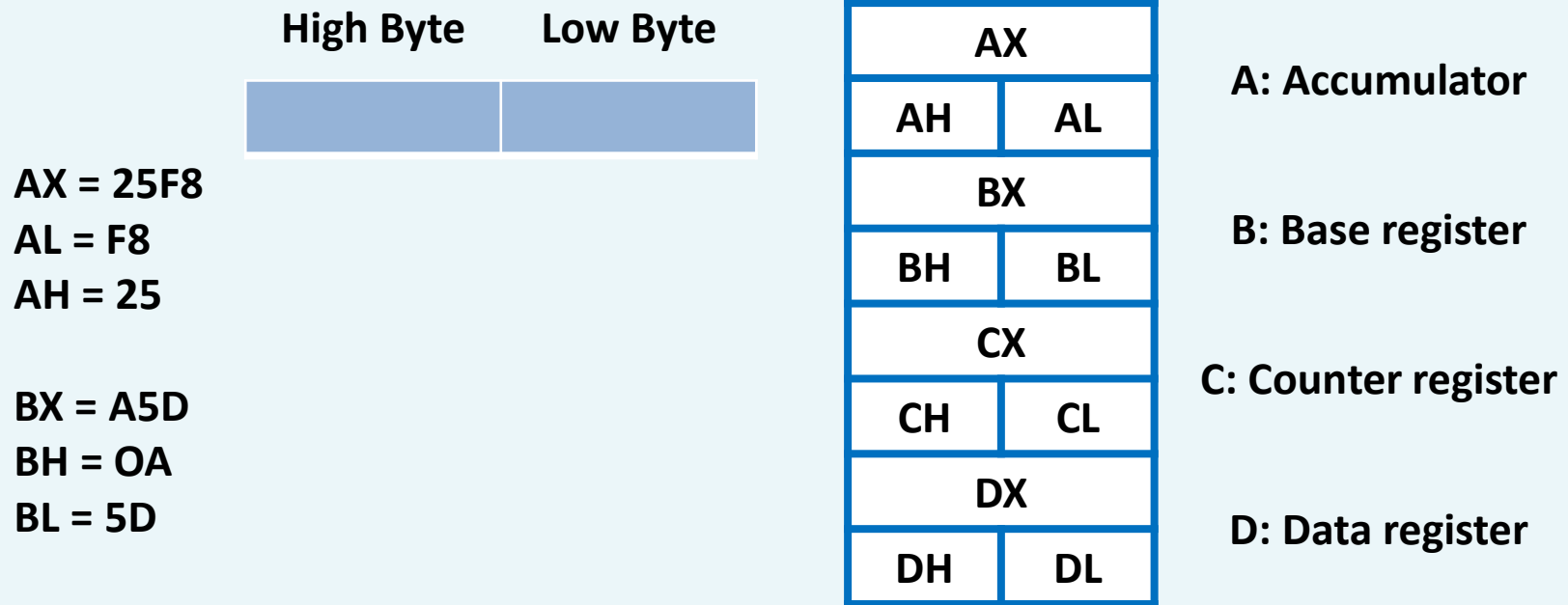
CX Register (Count register)

Used for controlling the number of times a loop is repeated, contains the value by which bits are shifted, it's also used for Computation.

DX Register (Data register)

Used for all Input/ Output operations. It is used for multiply and divide operations.

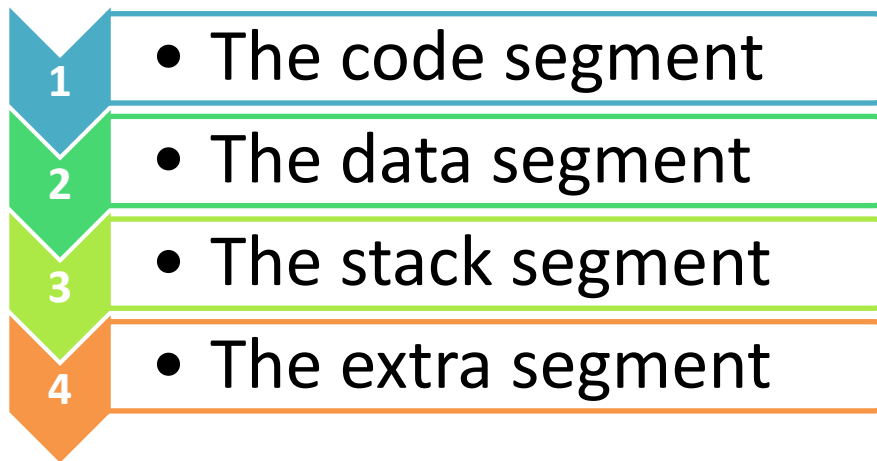
General Purpose Registers (Data Registers) (AX, BX, CX, and DX)



- Each register can be accessed as a Byte or a word.
- The left most byte is a high-order 8 bits of register and the right most byte is low-order 8 bits.

Segment Registers and Memory segmentation

- Even though the 8086 has a (1 M byte) address space, not all this memory can be active at one time.
- The (1 M bytes) of memory can be partitioned into 64 k byte (65536) segments.
- A segment: represents an independently addressable unit of memory consisting of 64k consecutive byte-wide storage location.
- Each segment is assigned a base address that identifies its starting point (its lowest address byte-storage location).
- Only four segments can be active at a time.



Size of segment = 64 byte

$$2^6 * 2^{10} = 2^{16}$$

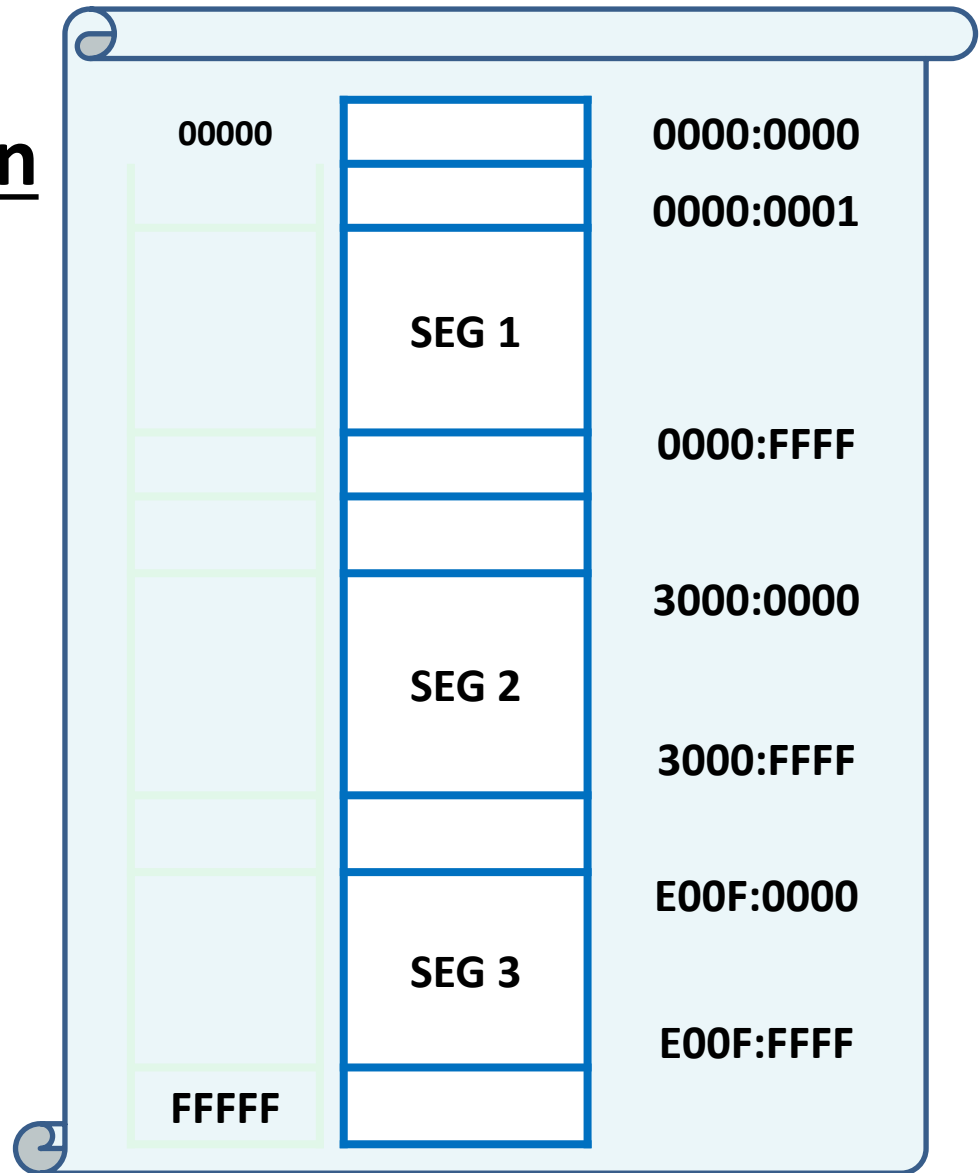
Size of address in segment

=16 bits

0000

FFFF

Memory Segmentation



Segment Registers and Memory segmentation

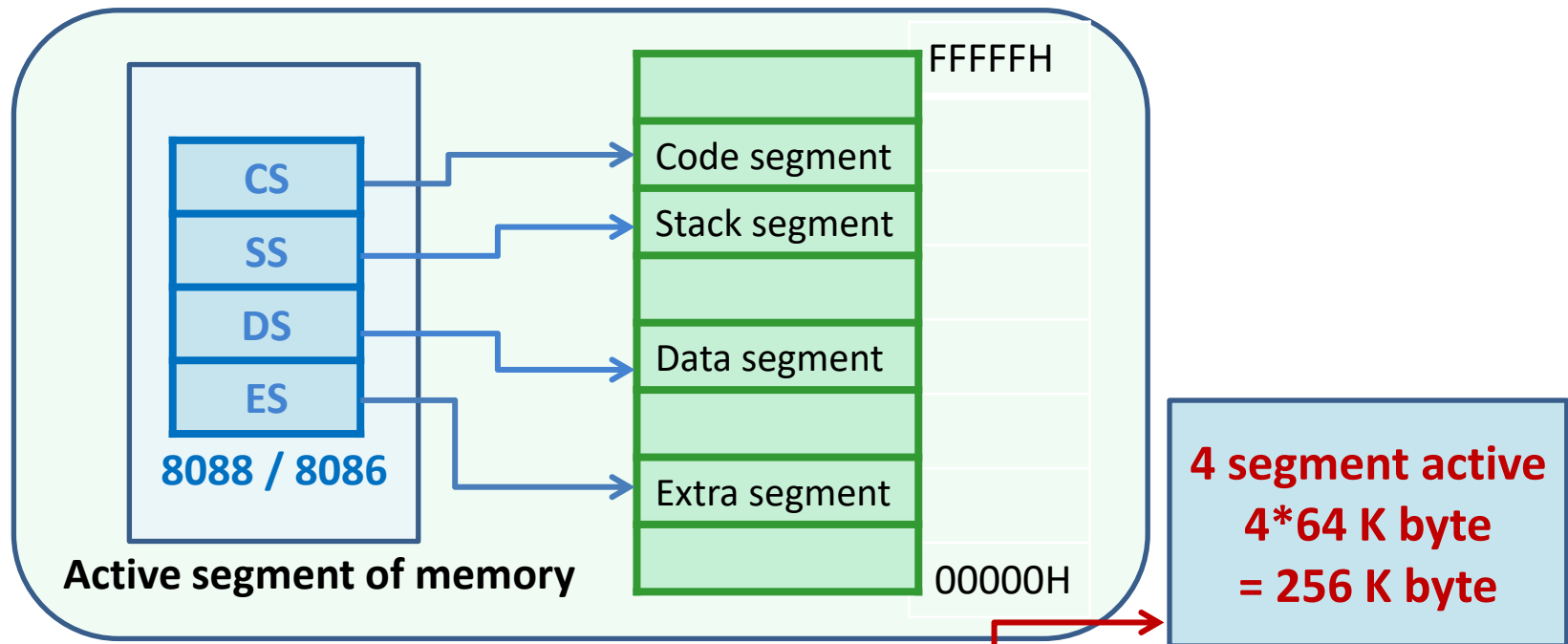
- Code Segments (CS) : holds the program instruction codes.
- Data Segment (DS) : stores data for the program.
- Stack Segment (SS) : used to store interrupt and subroutine return address.
- Extra Segment (ES) : is an extra data segment.

The segments of memory that are active are identified by the values of addresses held in the 8086's four internal segment registers.

- Code Segment Register (CS)
- Data Segment Register (DS)
- Stack Segment Register (SS)
- Extra Segment Register (ES)

Each of these registers contains a **16-bit base address** that points to **the lowest addressed byte of the segment in memory** .

Segment Registers and Memory segmentation



Four segments give a maximum of 256 K bytes of active memory . {64 K bytes are for code (program storage), 64 K bytes are for stack, and 128 K bytes are for data storage}

Logical and Physical addresses

Three types of addresses:

Physical Address

20 bits actually
put on the
address bus
Range: from
00000 to FFFFF

Offset Address

A location
within a 64 K
byte segment
Range: from
0000 to FFFF

Logical Address

Consist of a
segment value
and offset
address

SEGMENT VALUE : OFFSET ADDRESS

- The segment base address and offset address are **16 bit quantities**.
- This is because all register and memory locations used in address calculation are always 16 bits long. However, the physical address that are used to access memory are 20 bit in length.